A dependent product of n-types over an arbitrary type is an n-type

The Univalent Foundations Program

hott-book:hlevels.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'thm:hlevel-prod' (source lines 171-174, Theorem 7.1.9); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own sentence announcing closure of n-types under limits, which is the framing under which the statement sits.

Editorial note. The retained counter values (chapter 7, section 1, theorem offset 8) make the retained theorem print as Theorem 7.1.9, matching the original book numbering.

Editorial note. The single external reference points at the contractibility of dependent products used in the base case; it is a navigation pointer only.

Editorial scope. One original result of the HoTT Book, counted once: dependent products of n-types are n-types (Theorem 7.1.9).

Allowed standard prerequisites (editorial).

- Function extensionality. A homotopy between two dependent functions yields an equality of the functions.. It is what identifies f = g with the type of pointwise equalities at source line 183.
- Preservation of n-types under equivalence. If X is equivalent to an n-type, so is X.. It is invoked at source line 183 after the path space has been replaced by a product of path spaces.
- Contractible dependent products. A dependent product of contractible types is contractible.. It is the base case n = -2 at source line 178.

Source reference. Original theorem 3.11.6: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1757–1757. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

More generally, n-types are closed under all *limits*.

Theorem 7.1.9. *Let $n \geq -2$, and let $A : \mathcal{U}$ and $B : A \to \mathcal{U}$. If for all $a : A$, $B(a)$ is an n-type, then so is $\prod_{(x:A)} B(x)$.*

Proof. We proceed by induction on n. For $n = -2$, the result is simply original theorem 3.11.6.

For the inductive step, assume the result is true for n-types, and that each $B(a)$ is an $(n+1)$-type. Let $f, g : \prod_{(a:A)} B(a)$. We need to show that $f = g$ is an n-type. By function extensionality and closure of n-types under equivalence, it suffices to show that $\prod_{(a:A)}(f(a) =_{B(a)} g(a))$ is an n-type. This follows from the inductive hypothesis. □